

Adaptive Kernel Selection for Stein Variational Gradient Descent

Anonymous Authors

Abstract

Efficient posterior approximation remains a central problem in Bayesian inference. Stein Variational Gradient Descent (SVGD) addresses this problem by transporting an ensemble of particles toward a target distribution through update directions defined in a reproducing kernel Hilbert space (RKHS). Because the RKHS is determined by the chosen kernel, the kernel specification directly controls the interaction among particles and therefore has a decisive effect on convergence and approximation quality. The standard median heuristic provides a convenient bandwidth rule, but its limited flexibility can lead to poor approximations, especially in higher-dimensional settings. We develop Adaptive SVGD (Ad-SVGD), a principled kernel-selection strategy for parameterized kernel families that alternates between SVGD particle transport and gradient-ascent updates of the kernel parameters obtained by maximizing the kernelized Stein discrepancy (KSD). This construction turns recent KSD-based convergence insights into an explicit adaptive inference procedure. We give a simplified mean-field analysis that extends fixed-kernel KSD descent results to the adaptive setting and establishes convergence of the maximal KSD over the kernel class. Across the numerical problems studied here, Ad-SVGD improves on standard bandwidth heuristics and more accurately represents posterior uncertainty.

1 Introduction

Stein Variational Gradient Descent (SVGD) [SVGD] is a deterministic particle-based method for approximate Bayesian inference and a widely used alternative to traditional Markov Chain Monte Carlo (MCMC) methods. SVGD evolves an ensemble of particles along update directions induced by the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, with the admissible directions restricted to the unit ball of a reproducing kernel Hilbert space (RKHS). This RKHS restriction is also the source of a fundamental limitation: the kernel determines the geometry of the particle interaction and can substantially alter the performance of the algorithm [Duncan2019OnTG, Nsken2021SteinVG]. In practice, SVGD particle systems also often underestimate posterior variance [DBLP:conf/iclr/BaEGSSWZ22], a phenomenon that has reinforced the view that the method deteriorates as the dimension of the state space grows. This paper shows that a substantial part of this limitation can be addressed within the original SVGD framework by replacing static bandwidth selection with an adaptive kernel-parameter mechanism. By dynamically tuning the kernel through maximization of the kernelized Stein discrepancy (KSD), the proposed method selects update geometries that support more effective transport in the complex and higher-dimensional settings considered here.

1.1 Related work.

Since its introduction [SVGD], SVGD has become an established tool for approximate Bayesian inference across machine-learning applications [DBLP:conf/uai/LiuRLP17, DBLP:conf/iclr/MessaoudMXP

DBLP:conf/nips/PuGHLHC17, **DBLP:journals/tsp/KassabS22**]. A substantial theoretical literature now characterizes its behavior: mean-field convergence has been studied in continuous time [**LLN2019**, **Duncan2019OnTG**, **NEURIPS2020'16f8e136**] and discrete time [**NEURIPS20203202111**, **pmlr-v162-salim22a**], finite-particle convergence rates have been obtained under several sets of assumptions [**balasubramanian2024improvedfiniteparticleconvergencerates**, **NEURIPS2023'54e5d7af**], and the dynamics have been interpreted as gradient flows on spaces of probability measures [**DBLP:conf/nips/LiuDuncan2019OnTG**], in analogy with **VariationalFormulationFokker**.

Kernel choice is central in both the empirical and theoretical behavior of SVGD, because it specifies the particle interaction and enters directly into the convergence criterion (see also Figures 1 and 9). Existing convergence statements are commonly formulated in terms of the KSD, and the extent to which KSD convergence implies weak convergence depends on the selected kernel [**pmlr-v70-gorham17a**]. The median heuristic [**JMLR:v13:gretton12a**] remains attractive because of its simplicity, but it has no general theoretical justification in this setting and its performance can degrade as dimensionality increases [**DBLP:conf/iclr/BaEGSSWZ22**, **DBLP:conf/icml/ZhuoLSZCZ18**, **DBLP:conf/icml/WangZ018**]. Several SVGD variants have therefore been proposed to counteract high-dimensional performance degradation [**stein'newton**, **NEURIPS2019'5dcd0ddd**, **gong2021sliced**, **pmlr-v151-liu22a**], and related approaches learn update directions with neural networks [**langosco2022neural**, **ZHAO2023118975**].

The closest precedent is the mixture-of-kernels framework of **ai'stein'2023**. That method defines a KSD for a weighted linear combination of base kernels and learns the weights by maximizing the resulting multiple-kernel KSD. Our work retains the KSD-maximization principle while moving beyond finite kernel bases: we optimize continuous kernel parameters, including dimension-dependent bandwidths, thereby treating kernel adaptation as an integral component of SVGD rather than as a fixed finite mixture selection problem.

1.2 Contributions.

This work asks whether the original SVGD transport mechanism, without additional architectural or algorithmic modifications, can deliver competitive posterior approximations when equipped with a principled adaptive kernel. By isolating kernel selection from the broader family of SVGD enhancements, we identify the extent to which the kernel alone governs the effectiveness of the method and show how adaptive kernel learning can substantially improve the behavior of the original algorithm.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We introduce Ad-SVGd, which updates kernel parameters during inference by ascending the KSD before applying the SVGD particle update. Unlike the median heuristic, which fixes a scalar bandwidth rule, the method accommodates multiple continuous kernel parameters and therefore supports anisotropic and iteration-dependent update geometries.
- (ii) **Theoretical Analysis.** We analyze the algorithm in the discrete-time mean-field setting and extend fixed-kernel convergence results for SVGD to an adaptive kernel class. In particular, we prove convergence of the supremal KSD over the parameterized class and, under a Stein logarithmic Sobolev inequality, derive mean-field iteration-complexity bounds.

- (iii) **Empirical Validation.** We demonstrate on Gaussian mixtures, Gaussian targets of increasing dimension, inverse problems, Gaussian process inference, and Bayesian logistic regression that adaptive kernel selection improves the approximations obtained with the median heuristic and mitigates the variance collapse observed in the baseline.

2 Mathematical background

We fix a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and let $\mathcal{H}_0$ denote its associated RKHS. The vector-valued RKHS used by SVGD is the d -fold Cartesian product $\mathcal{H}$, equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. We write $x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted by $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ denotes the subset of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ whose marginals are μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [IPM] quantify the discrepancy between two measures by taking the largest difference of expectations over a prescribed test-function class. Stein’s method [bounderrornormal] makes this construction useful for intractable targets by producing test functions whose expectation under the target distribution vanishes. For the Stein operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [KernelTestGoodness, KernelizedSteinDiscrepancy]. The associated *kernelized Stein discrepancy* (KSD) is therefore defined by

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. The optimizer is $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$ and Φ_k is the feature map associated with k . Consequently, the supremum equals the RKHS norm of the Stein witness function, yielding the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and an initial distribution μ_0 , SVGD constructs an approximation of π through the push-forward recursion $\mu_{n+1} := T_{\#} \mu_n$, where $T_{\#} \cdot$ denotes the push-forward by $T(x) = x + \gamma \psi^{\mu_n}(x)$ and $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ is the RKHS steepest-descent direction. This update follows from the solution of Equation (1): $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes the directional derivative $\frac{d}{d\gamma} \text{KL}(T_{\#} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [SVGD], where $\text{KL}(\cdot \| \cdot)$ denotes the KL-divergence. For this choice,

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\#} \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying these transport maps to particles $\{X_0^i\}_{i=1}^M$ sampled from μ_0 gives the finite-particle SVGD recursion

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because KL convergence for SVGD is difficult to prove without restrictive assumptions, recent analyses focus on convergence in KSD [NEURIPS20203202111c, NEURIPS2023*54e5d7af, pmlr-v162-salim22a]. KSD convergence alone, however, does not automatically imply weak convergence: the associated mean-field measures may fail to be tight, which is necessary for weak convergence on Polish spaces. Moreover, Equation (1) makes explicit that the RKHS—and hence the kernel—determines how informative KSD convergence is about convergence of probability measures [pmlr-v70-gorham17a].

We therefore formulate SVGD over a parameterized kernel family $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, let KSD_θ denote the corresponding *kernelized Stein discrepancy*, ψ_θ^μ the associated optimal update direction, and Φ_θ the corresponding feature map. This parameterization makes kernel choice part of the optimization procedure and allows the algorithm to adapt the RKHS geometry to the current particle approximation instead of committing to a fixed kernel throughout inference.

The adaptive rule is based on a direct variational principle: at the current particle measure, select the kernel that maximizes the KSD to the target. Since the instantaneous KL decrease under SVGD is proportional to the squared KSD (see Equations (2) and (7)), maximizing the KSD selects, within the prescribed kernel class, the update geometry that gives the largest first-order decrease. This interpretation is consistent with geometric analyses of SVGD as a gradient flow, where kernels producing larger KSD values yield more favorable convergence behavior when the corresponding RKHSs are compared [Nskn2021SteinVG, Duncan2019OnTG].

Whereas standard SVGD usually studies KSD convergence for a fixed kernel, Ad-SVGd targets the largest KSD attainable within the chosen kernel class. In contrast to the median heuristic used by **SVGD**, the mechanism applies to any parameterized kernel family for which the KSD can be evaluated and differentiated: at iteration n , the idealized objective is

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

The practical algorithm approximates this maximization by performing one or more gradient-ascent steps with step size $s > 0$ on the KSD before the particle update, and it permits kernel updates to be skipped on selected SVGD iterations to reduce runtime (see Algorithm 1). This option is controlled by the user-specified decision variable `paramupdate`. The particle-transport step itself can be replaced by standard SVGD refinements, such as adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates, without changing the kernel-selection mechanism. Likewise, the gradient-ascent update for the kernel parameter can be substituted by another iterative optimizer. Our implementation computes the kernel-parameter gradient from the explicit KSD representation [**KernelTestGoodness**, **KernelizedSteinDiscrepancy**]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k \, d(\mu \otimes \mu) \quad (4)$$

with

$$u_{\pi}^k(x, y) = k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \quad (5)$$

which permits direct differentiation with respect to the kernel parameters. The additional cost relative to vanilla SVGD is limited by two implementation facts:

- The gradient-ascent steps for the kernel parameter reuse the same score evaluations $\nabla \log \pi(X_n^i)$ as the corresponding SVGD step, so Ad-SVGd does not require additional evaluations of $\nabla \log \pi$.
- The cost can be controlled by taking a small number of kernel-parameter steps nstepstheta , by updating the kernel only after several particle updates, or, for large particle ensembles, by estimating the KSD with a particle subsample.

Algorithm 1: Ad-SVGd

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_{\theta} \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps nsteps , $\text{nstepstheta} \in \mathbb{N}$

Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to  $\text{nsteps} - 1$  do
    Decide  $\text{paramupdate} \in \{\text{True}, \text{False}\}$ ;
    if  $\text{paramupdate}$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $\text{nstepstheta} - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^{\ell} + s \nabla_{\theta_n^{\ell}} \text{KSD}_{\theta_n^{\ell}}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We consider a target measure with Lebesgue density of the form $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. To better motivate our proposed Ad-SVGd, we will demonstrate how to extend recent results on the convergence of SVGD in the sense of KSD. To be more precise, we will extend the convergence analysis conducted in [pmlr-v162-salim22a] under the Talagrand’s inequality which holds under mild assumptions on the target distribution and is weaker than the commonly employed logarithmic Sobolev inequality. We make the same assumptions as in [pmlr-v162-salim22a] uniform over all kernel parameters.

Assumption 1. We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.

Assumption 2. We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand's inequality T1, which means that there exists $\lambda > 0$ such that

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \| \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand's inequality

In the following, we denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and make use of the following fundamental inequality. Given a fixed kernel parameter $\theta \in \Theta$ such that Assumption 3 is satisfied, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$ and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Then, according to Proposition 3.1 in [pmlr-v162-salim22a], it holds

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) with a fixed kernel parameter $\theta \in \Theta$ one can then derive the descent condition [pmlr-v162-salim22a],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The key aspect to verify (7) is the verification of $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ using Talagrand's inequality, Assumption 2, which allows to apply (6) for $g = \psi_\theta^{\mu_n}$. Condition (7) can then be used to argue that

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

since $\sum_{n=0}^{\infty} \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) > 0$.

When introducing an adaptive kernel parameter choice (θ_n) , the inequality (7) changes to

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

Suppose the adaptive SVGd iteration can be written in the following simplified form

$$\begin{aligned}\mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),\end{aligned}\tag{9}$$

meaning that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu_n | \pi).$$

We emphasize that this formulation is possible when the maximization of the KSD with respect to the kernel parameter has a (unique) closed-form solution. This is actually the case in the multiple-kernel SVGd framework [ai'stein'2023], which parameterizes the kernel as a convex combination of base kernels

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For the general setting, the kernel parameter update θ_n in (9) needs to be approximated. Note that in the general setup we assume that $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition is a direct consequence of Theorem 3.2 in [pmlr-v162-salim22a]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) = 0$.*

The situation changes when we assume that we can only approximately solve the maximization task of the KSD with respect to the kernel parameter. Suppose that $\max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned}\theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n\end{aligned}\tag{10}$$

for some sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ with the goal to maximize $\operatorname{KSD}_{\theta}(\mu_n | \pi)$. Specifically, in Algorithm 1 Ψ_n corresponds to the gradient ascent update for the KSD. However, as mentioned above, this update could be replaced by other suitable iterative optimization schemes. Our required assumption on the update rule is the following convergence behavior.

Assumption 4. We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Using this assumption, we can make the following convergence guarantee.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [pmlr-v162-salim22a] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under the so-called Stein logarithmic Sobolev inequality, we can further extend the convergence of Ad-SVGD to convergence in the KL divergence. Under specific scenarios for ε_n we derive iteration complexity bounds for the mean-field limiting system (10). Moreover, this setting allows us to explicitly describe the bias of Ad-SVGD when the KSD maximization is only solved up to a fixed $\bar{\varepsilon}$ -accuracy in each iteration. The details can be found in Appendix B. Finally, we emphasize that we require only one kernel in our kernel class $\{k_{\theta}\}$ to satisfy the Stein logarithmic Sobolev inequality which yields a promising flexibility to verify this condition.

5 Numerical Experiments

In the following section, we evaluate Ad-SVGD on two numerical experiments: a one-dimensional Gaussian mixture model and a linear inverse problem governed by an ordinary differential equation. Additional toy experiments examining the effect of increasing dimensionality are provided in Appendix A. Moreover, in Section 5.4 we demonstrate the advantages of Ad-SVGD over SVGD with the median heuristic on a Bayesian logistic regression model, which was also used in prior SVGD literature.

5.1 Kernel parameterization

SVGD is most commonly used with kernels of the form

$$k_h(x, y) = \exp \left(-\frac{\|x - y\|_p^p}{h} \right),$$

where $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [SVGD, DBLP:conf/iclr/BaEGSSWZ22, Duncan2019OnTG]. We will focus on selection strategies for the parameter h , which is known as the kernel *bandwidth*. The commonly used heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med denotes the current median distance between the particles. This choice is motivated by the goal of balancing the two terms in the SVGD update (3) [SVGD].

To take advantage of the flexibility of our adaptive method, we use product kernels of the form

$$k_h(x, y) = \prod_{i=1}^d \exp \left(-\frac{|x_i - y_i|^p}{h_i} \right)$$

with parameter $h = (h_1, \dots, h_d)$, i.e. we allow for dimension-dependent bandwidths. The derivatives necessary to apply Algorithm 1 with these kernels (see Equations (4) and (5)) can easily be calculated. We also tested using an adjusted version of the median heuristic with these kernels taking a naive median for each dimension. However, this approach did not produce good results and suffered from a variance collapse.

Beyond our current experiments, the flexibility of Ad-SVGD allows for more scalable kernel families in higher-dimensional settings. In particular, spectral or low-rank kernel parametrizations can adapt the dominant directions of variability without incurring the exponential cost of full product kernels, while Matérn or mixed-product constructions provide additional control over smoothness and anisotropy.

5.2 Toy example

We first consider the one-dimensional example from SVGD. This is a Gaussian mixture with two components: $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. We use $M \in \{50, 200, 500\}$ particles and sample the initial particle set from $\mathcal{N}(0, 1)$. We run SVGD for 10^4 steps with a step size of 1, using kernels of the form described above with $p = 1$ and different choices of (fixed) bandwidth h . As a measure of sample quality, we use the Wasserstein 1-distance $\mathcal{W}_1$, which we compute using an implementation of the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see StatisticalAspectsWasserstein) and an exact sample of size 10^5 . The results of these experiments are shown in Figure 1, where we see that the algorithm performs well only for bandwidths within a certain range. The algorithm is highly sensitive to the choice of the parameter h and therefore, a careful selection strategy is crucial.

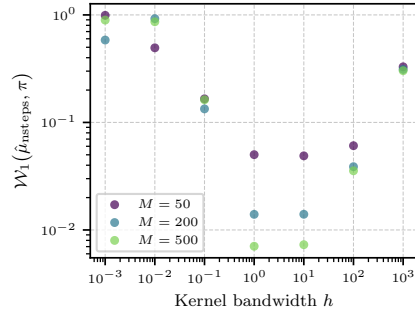


Figure 1: Final Wasserstein 1-distances for one-dimensional example using SVGD with different fixed bandwidths h .

5.3 Linear inverse problem based on ODE

The following example is adapted from [Weissmann2022Optimization]. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observation points of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. These observations are described by

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is defined by $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ being the observation operator at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation of the inverse problem, we consider a Gaussian process (GP) prior for u given by the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$. The resulting problem is to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ with prior given by $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Assuming $\varepsilon \sim \mathcal{N}(0, \Gamma)$, this leads to the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For the implementation, we replace H by a numerical discretization operator for Equation (11) using a grid with mesh size 2^{-8} and consider the fully observed system (i.e. $N_{\text{obs}} = 2^8$). We use $N_x = 16$ terms for the KL expansion of u and assume noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$. We compare the performance of SVGD using the median heuristic (we call this Med-SVGD) with our Ad-SVGD for different choices of particle ensemble size $M \in \{50, 100, 200\}$. We use the kernels described in Section 5.1 with the choice $p = 1$. We run both algorithms for $4 \cdot 10^5$ iterations using the step size 10^{-3} for particle updates and, as suggested by **SVGD**, a variant of AdaGrad for adaptive step size control. In Ad-SVGD we use the step size 10^{-5} for the bandwidth updates and update the bandwidth only once for every 100 particle updates. With this update scheme, there was no significant runtime difference between Med-SVGD and Ad-SVGD.

Figure 2 shows the GP reconstruction for an exemplary seed. We observe that both methods are able to give a good approximation of the mean, but only Ad-SVGD correctly captures the posterior

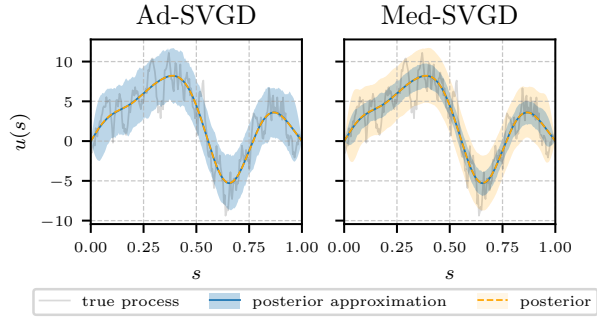
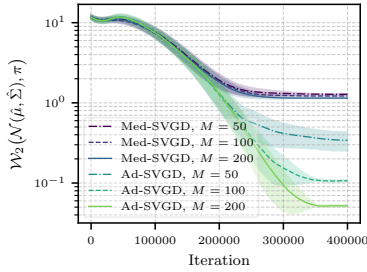


Figure 2: GP reconstruction for ODE-based inverse problem, showing mean and 90% confidence interval

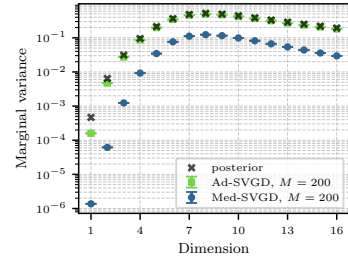
uncertainty. To further quantify the approximation quality, we use the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the normal distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ is the sample mean and $\hat{\Sigma}$ the sample covariance of the particle set [StatisticalAspectsWasserstein]. Since the target π is a multivariate normal distribution, this has an explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the mean and covariance of the posterior. We also compare the marginal variances of the final particle distribution with the posterior. Figure 3 shows the results of these experiments aggregated over 56 different random seeds (note that the posterior covariance does not actually depend on $\bar{y}$). Again, we observe that Ad-SVGD achieves better approximations of the posterior than Med-SVGD, which underestimates the uncertainty of the problem. Furthermore, in contrast to Med-SVGD, the approximation quality of Ad-SVGD improves as the number of particles increases beyond 50. Figure 4 shows the behavior of the bandwidths determined using Ad-SVGD. We observe that the component-wise bandwidths stabilize more quickly than the approximation error. Clear differences across the components are visible, with the final bandwidths being negatively correlated with the corresponding marginal variances.

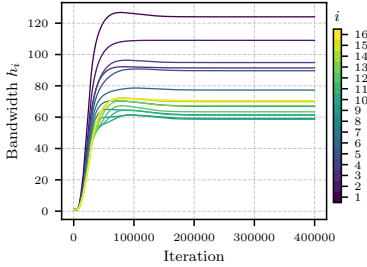


(a) Convergence of approximation error

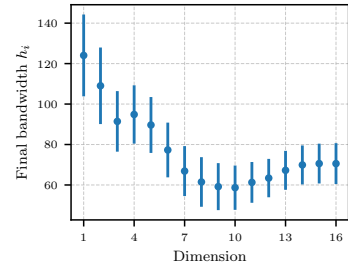


(b) Marginal variances

Figure 3: Aggregated results (mean and 95% confidence interval over 56 random seeds) for ODE-based inverse problem using Med-SVGD and Ad-SVGD



(a) Evolution of bandwidth per dimension (mean)



(b) Final bandwidths (mean and 95% conf. interval)

Figure 4: Behavior of bandwidth parameter for ODE-based inverse problem using Ad-SVGD, aggregated over 56 random seeds

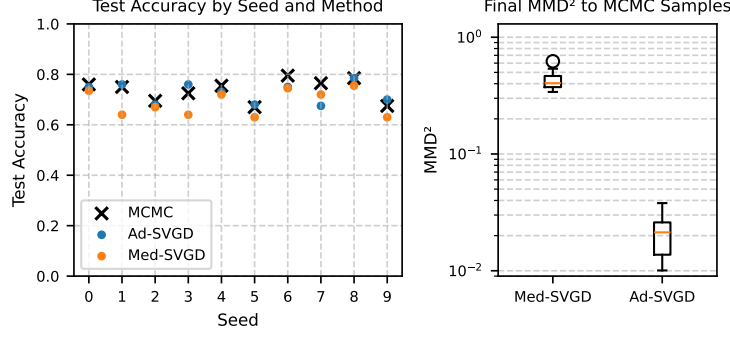


Figure 5: Approximation quality of SVGD particles for Bayesian logistic regression measured by prediction accuracy (left) and MMD^2 to reference samples (right)

5.4 Bayesian logistic regression

As in **SVGD** and **pmlr-v151-liu22a**, we consider a Bayesian logistic regression (BLR) model applied to the Covertypes data set [covertypes]. For the regression weights w , we assign a Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$ and we want to infer the posterior of $x = [w, \alpha]$. Following **pmlr-v151-liu22a**, we use the No-U-Turn Sampler (NUTS) introduced by **nuts** to generate a reference for evaluating the posterior approximation quality of Med-SVGd and Ad-SVGd. Again following **pmlr-v151-liu22a**, we test the methods on subsets of the original data set of size 1000 and aggregate our results over 10 random draws. To evaluate the performance of the methods for the classification task, we compute the prediction accuracy over a test set achieved using the particle mean. As shown in Figure 5, Ad-SVGd outperforms Med-SVGd across almost all seeds, achieving a higher test accuracy while also being closer to the MCMC reference. To measure the accuracy of the posterior approximation, the same figure also shows the squared maximum mean discrepancy (MMD^2) [**MMDref**] to the MCMC samples. We observe a significant difference of how well the two methods are able to capture the posterior distribution. This is further confirmed by Figure 7, which shows the covariance matrices of the final particles compared to the MCMC reference, averaged over the 10 seeds (this visualization style is taken from **pmlr-v151-liu22a**). We observe that Med-SVGd severely underestimates the posterior uncertainty, while Ad-SVGd gives a better approximation. Additionally, Figure 6 shows the evolution of the bandwidths for each component. We see that from an uninformed initialization (1 in every component), Ad-SVGd is able to recover suitable bandwidths, which differ from the initialization by several orders of magnitude.

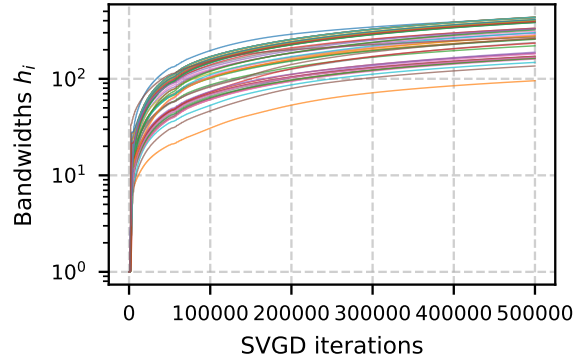


Figure 6: Bandwidth evolution for BLR example

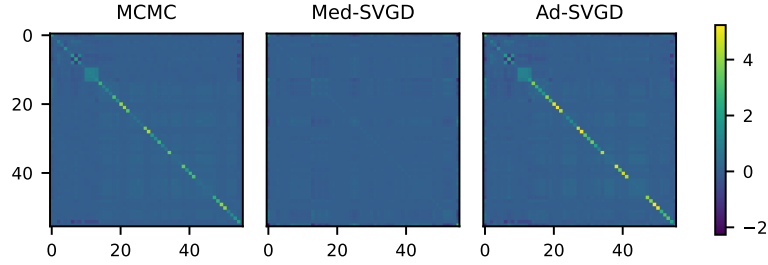


Figure 7: Covariance matrices of final SVGD particle sets compared to MCMC samples (mean over 10 seeds)

6 Limitations and outlook

The main limitation of our analysis is its reliance on Assumption 4. In our ongoing work, we examine when Assumption 4 is satisfied by the alternating gradient-ascent scheme in Algorithm 1 used in our experiments. Although we have no guarantees for Assumption 4 to be satisfied, our implementation led to promising empirical results.

Our considered analysis focused on the original dynamic of the SVGD, and it would be intriguing to combine the proposed adaptive kernel selection with recent variants such as sliced SVGD [gong2021sliced], Grassmann SVGD [pmlr-v151-liu22a], or Stein transport [nsken2024steintransportbayesi]

A More numerical experiments

A.1 Gaussian mixture models

We revisit the example from Section 5.2. For the setup described there, Figure 9 shows histograms of the final particle distribution in comparison to the target density for $h \in \{0.001, 1, 1000\}$. We now turn to the comparison of Med-SVGD with Ad-SVGD. We consider different numbers of particles M and compare the final particle distributions after 10^4 iterations with step size 1 for the two methods.

Figure 8 shows the Wasserstein 1-distance between the final empirical distribution of the particles and the target distribution using Med-SVGD and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. We observe that, as expected, the approximation quality improves with N for both methods. Both methods achieve similar results, reaching a Wasserstein distance below 0.01 for $N = 500$.

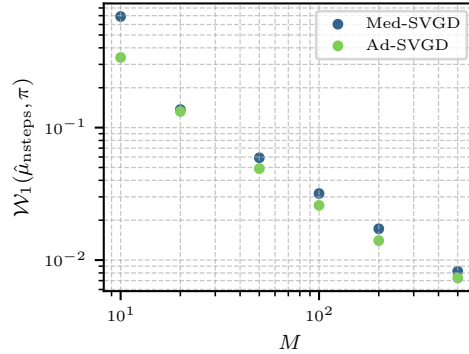


Figure 8: Approximation error for 1D Gaussian mixture.

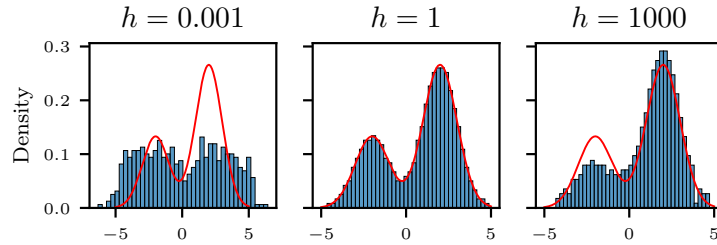


Figure 9: Histograms of final particle set generated using fixed bandwidths $h = 0.001, 1, 1000$; target density π shown in red for comparison.

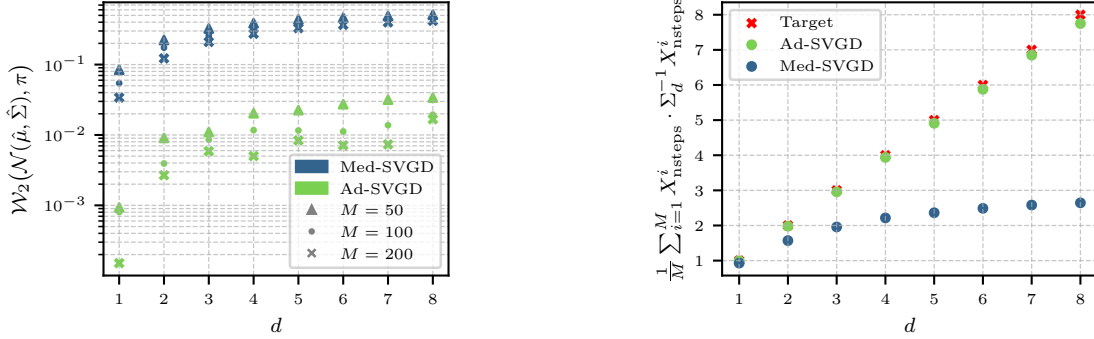
In the one-dimensional case, both methods are able to approximate the target distribution well. Our adaptive bandwidth selection strategy works well, but has no significant advantage over the commonly used median heuristic in this scenario.

A.2 Scaling dimension: multivariate normal distribution

Moving into higher dimensions, we now consider the Gaussian target distributions $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. We used $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ as the initial particle distribution and ran the algorithms for 10^4 iterations with step size 0.1. Whenever necessary for numerical stability, we used a smaller step size and adjusted the number of iterations accordingly.

We compare the approximation quality of the final set of particles generated using Med-SVGD and Ad-SVGD as the dimensionality of the problem increases. Because the Wasserstein 2-distance

between two Gaussian distributions has an explicit formula (see [StatisticalAspectsWasserstein]), we use as a measure of sample quality the Wasserstein 2-distance between the target distribution π_d and the Gaussian distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance matrix calculated from the set of particles [StatisticalAspectsWasserstein]. In accordance with the corresponding function of the Python Optimal Transport library [PythonOptimalTransport], which we used for our calculations, we call this the *Bures Wasserstein distance*. Figure 10a shows the development of this sample quality measure, achieved using Med-SVGD and Ad-SVGD with $M = 50, 100, 200$, as the dimensionality of the problem increases. We see that Ad-SVGD significantly outperforms Med-SVGD for all values of d .



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic for $M = 200$.

Figure 10: Results for multivariate normal distributions of increasing dimension

For the Bures Wasserstein distance, the smaller variances in the last components of our target distributions π_d do not have a big impact; to see the impact of failing to correctly capture the uncertainty in these components, we consider the test statistic $X \cdot \Sigma_d^{-1} X$ which is χ^2 -distributed with d degrees of freedom (i.e. it has expected value d) for $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [Metropolisadjustedinteracting]). We calculate the mean of this statistic on the set of particles with $M = 200$ and compare it to the expected value d in Figure 10b. Moving beyond the one-dimensional case, the test statistic for Med-SVGD deviates further and further from the true expected value as d increases, while the statistic for Ad-SVGD stays relatively close to the true expectation. This shows the failure of Med-SVGD to correctly approximate higher-dimensional distributions, while Ad-SVGD is able to deal with those examples well.

We provide more details about the final particle distributions for $M = 200$ and different d in Tables 1 and 2. They list, for each d , the marginal variances of the set of particles compared to the marginal variances of the target distribution, which are given in the first row. Table 1 shows the results for Med-SVGD, Table 2 shows the results for Ad-SVGD. We see that Ad-SVGD is able to achieve a good approximation of the target distribution in all components in terms of the marginal variances. The particles generated using Med-SVGD, on the other hand, significantly underestimate the uncertainty of the target distribution. This matches the observations already made in Figure 10b.

Lastly, we focus on the marginal particle distributions for $M = 200$ and $d = 8$. To ease the visualization, we normalized them by scaling each component of the particles with the inverse of the corresponding marginal standard deviation of the target distribution (i.e. we multiplied the i -th component with i). This turns each marginal distribution of π_d into a standard normal distribution.

Table 1: Marginal variances of final particle distribution for d -dimensional examples generated using Med-SVGD with $M = 200$; marginal variances of the target distribution π_d shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of final particle distribution for d -dimensional examples generated using Ad-SVGD with $M = 200$; marginal variances of the target distribution π_d shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

Figures 11 and 12 show the histograms of these normalized marginal particles distributions for $d = 8$ generated using Med-SVGD and Ad-SVGD, respectively. A standard Gaussian density is shown in each plot for comparison. Again, we observe that Ad-SVGD is able to capture all marginal distributions well, while Med-SVGD underestimates the uncertainty of the target distribution. These observations are also visible in Figures 13 and 14, where the quantiles of the normalized marginal particle distributions are compared against the target quantiles (i.e. against a standard normal distribution).

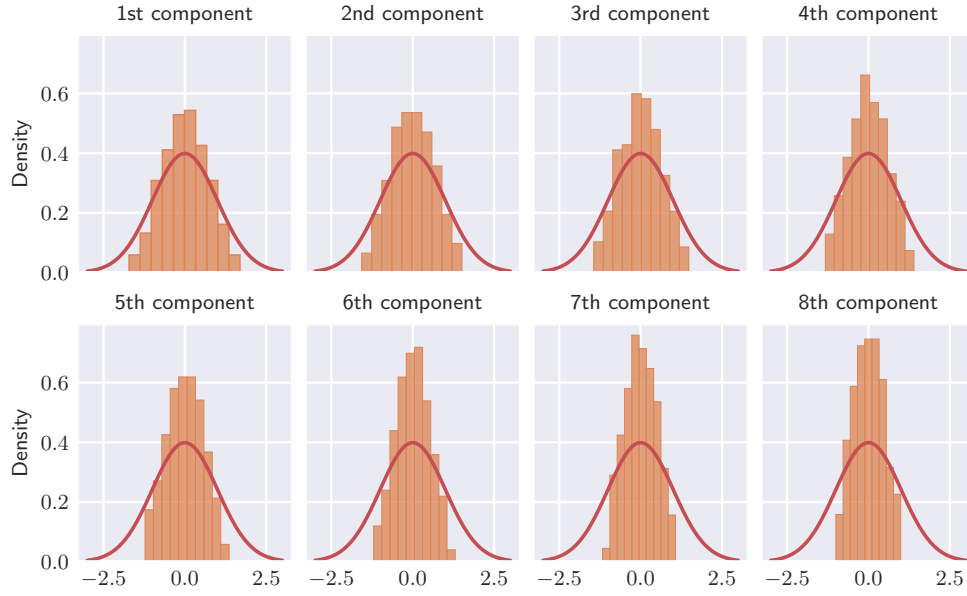


Figure 11: Histograms of single components of the final set of particles for eight-dimensional example generated using Med-SVGD with $M = 200$ and rescaled using marginal target variances; standard Gaussian density shown for comparison.

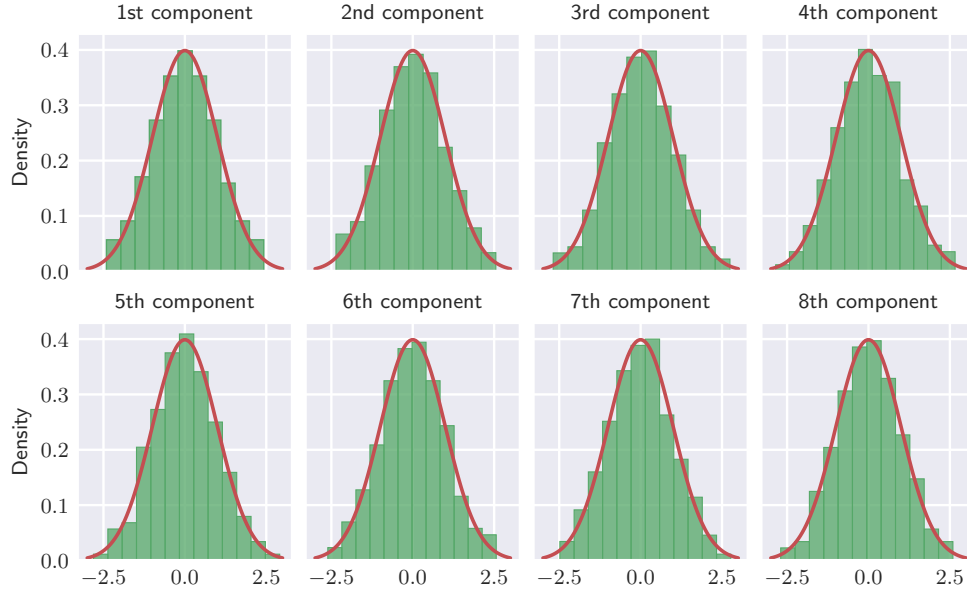


Figure 12: Histograms of single components of the final set of particles for eight-dimensional example generated using Ad-SVGD with $M = 200$ and rescaled using marginal target variances; standard Gaussian density shown for comparison.

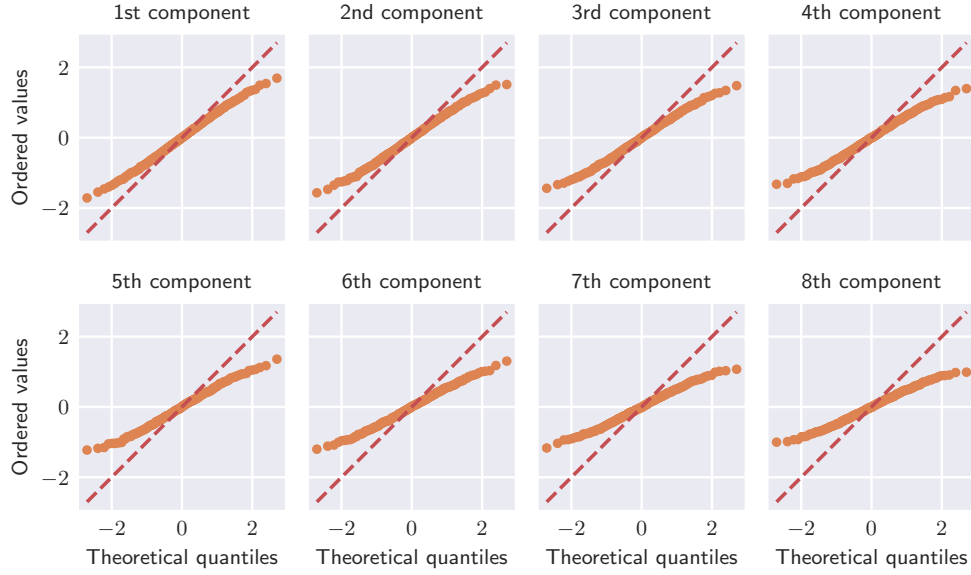


Figure 13: Q-Q plots comparing the marginals of the final particle distribution for eight-dimensional example generated using Med-SVGd with $M = 200$ and rescaled using marginal target variances with a standard normal distribution; line of slope 1 passing through the origin shown for comparison.

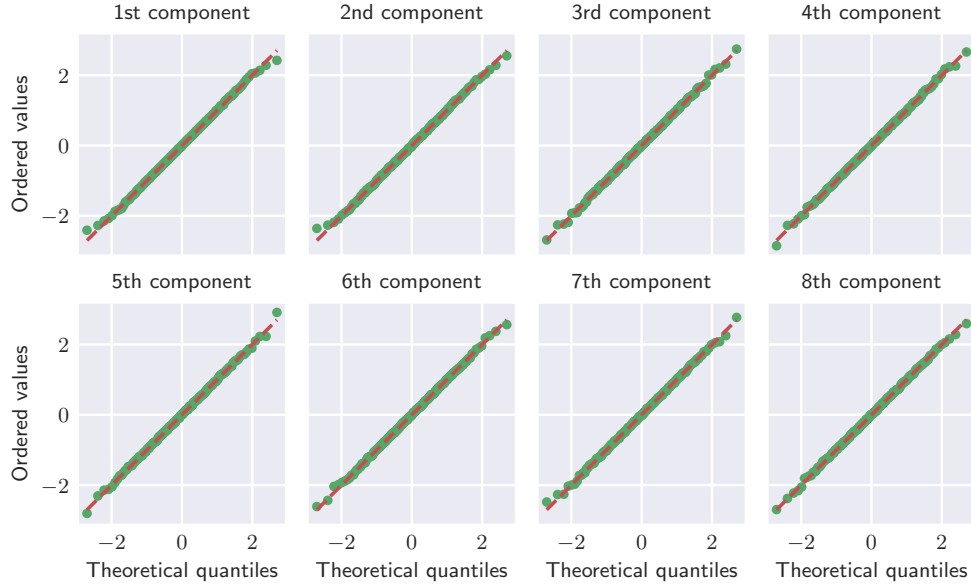


Figure 14: Q-Q plots comparing the marginals of the final particle distribution for eight-dimensional example generated using Ad-SVGd with $M = 200$ and rescaled using marginal target variances with a standard normal distribution; line of slope 1 passing through the origin shown for comparison.

A.3 Gaussian process inference

We demonstrate the behavior of our Ad-SVGD for scaling dimensions in an inference task of a Gaussian process (GP) proposed in [reich21]. We consider a GP on $[0, 1]$ represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ is a vector of independent Gaussian random variables $x_k \sim \mathcal{N}(0, k^{-2})$. We observe the process at N_y equispaced points in $[0, 1]$ and infer the coefficients x_k . For fixed N_x and N_y , this corresponds to an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal matrix Σ with entries k^{-2} , $k = 1, \dots, N_x$ and we assume independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_y}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

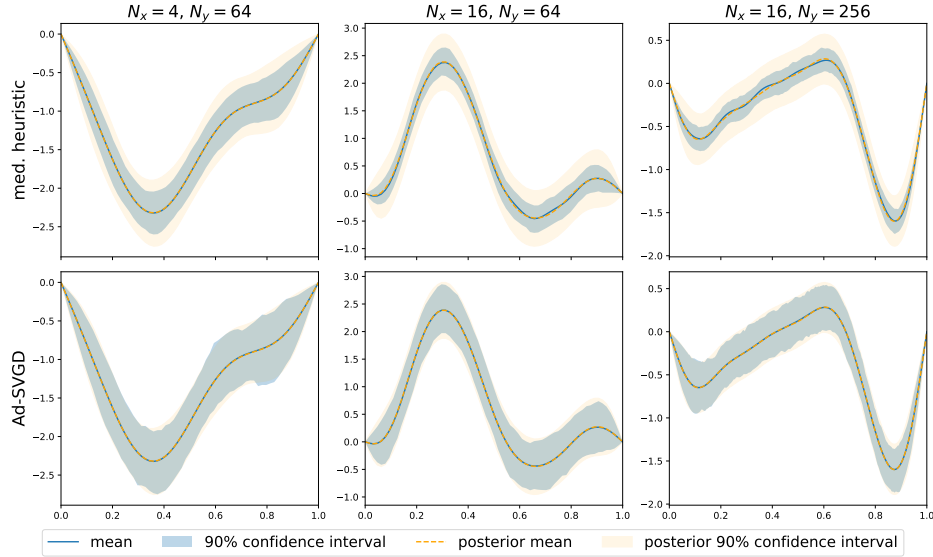


Figure 15: Estimated processes generated using the median heuristic and Ad-SVGD with $M = 100$ particles for different model configurations compared to posterior.

We use SVGD to sample from the resulting posterior $X | Y = \bar{y}$ and compare the performance of the median heuristic with our adaptive approach. For SVGD with the median heuristic, we use kernels of the form $k(x, y) = \exp(-\|x - y\|_1/h)$; for Ad-SVGD, we use product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with parameter $h = (h_1, \dots, h_{N_x})$, i.e. we use a different bandwidth for each dimension. Following [SVGD], we use a variant of Adagrad for step size control in the scenarios with $N_x = 16$.

Figure 15 shows the resulting processes in comparison to the posterior for the choices $N_x = 4$ and $N_y = 64$, $N_x = 16$ and $N_y = 64$ as well as $N_x = 16$ and $N_y = 256$. As the dimension increases, SVGD with the median heuristic underestimates the posterior variance, while Ad-SVGD is able to give a better approximation. The behavior is consistent across different numbers of observations of the Gaussian process. In Table 3, we quantify these results by comparing the trace of the covariance of the particle distributions generated by SVGD with the true posterior. SVGD with the median heuristic severely underestimates the uncertainty while Ad-SVGD is able to capture the variance more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of covariance of final particle distribution ($M = 100$ particles) compared to theoretical posterior for different configurations, averaged over 25 runs.

B Refined theoretical analysis

We can further improve our convergence result under a so-called Stein log-Sobolev inequality, which relates the KL divergence to the KSD. **Duncan2019OnTG** introduced this inequality and discussed conditions under which it might hold. Interestingly, in our adaptive setting of kernel selection, we can relax this condition by requiring only one kernel in our entire kernel class to satisfy the log-Sobolev inequality. More precisely, we make the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \mid \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Using this property, we can quantify convergence in terms of the KL divergence. We emphasize that in this scenario we can give an error bound on $\mathcal{F}(\mu_n)$ even under an approximate condition $\varepsilon_n \leq \bar{\varepsilon}$ without requiring $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□